

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch: Computer Science and Engineering/CE Semester: VI Subject Code & Name: BTCOC603_Y23 Machine learning Max Marks: 60 Date:20/01/24 2to5pm. Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/Co)	Marks
Q. 1	Solve Any Two of the following.		12
	a) Define Machine Learning and Enlist Future scope of Machine Learning with appropriate example?	B1/CO1	6
	b) Explain different types of learning using suitable real-world examples.	B3/CO1	6
	c) Explain Overfitting and methods to avoid Overfitting.	B2/CO1	6
Q. 2	Solve Any Two of the following.		12
	a) Explain Naïve Bayes classifier with its advantages and disadvantages?	B1/CO2	6
	b) Define kernel functions and Kernel SVM.	B2/CO2	6
	c) Define the following- i) Probability Density Function. ii) Probability Mass Function.	B3/CO2	6
Q. 3	Solve Any Two of the following.		12
	a) What is Perceptron, what are the basic Components of the Perceptron.	B6/CO3	6
	b) Define back propagation and error function.	B1/CO3	6
	c) What are Types of Deep Learning Networks and enlist applications for it.	B4/CO3	6
Q. 4	Solve Any Two of the following.		12
	a) Difference between PAC and PLC.	B2/CO4	6

	b) Explain Unrestricted hypothesis space and Restricted hypothesis space.	B5/CO4	6
	c) Justify Solvable vs. Unsolvable Problems with Computational theory.	B6/CO4	6
Q. 5	Solve Any Two of the following.		12
	a) How to choose value of “K number of clusters” in k-mean clustering.	B1/CO5	6
	b) Define dendrogram and approaches hierarchical clustering technique.	B4/CO5	6
	c) Define Gaussian Mixture Model (GMM) and Maximum Likelihood Estimation with respect with GMM?	B2/CO5	6
	*** End ***		